PHASE 2 • REQUIREMENT ANALYSIS

✓ VERIFIED SPECIFICATION

Technology Stack

AI-Powered Travel Planner • Smart Situational Itinerary Engine

Live URL: travel-planner-iota-plum.vercel.app **Repo:** github.com/itskian2007/travel-planner **Tech:** React + FastAPI + PostgreSQL **Team:** itskian2007

1. Architectural Tier Breakdown

The technology stack was selected to achieve three core engineering objectives: (1) **Sub-second generation latency**, (2) **Seamless cloud scalability**, and (3) **Zero infrastructure maintenance cost** using modern serverless edge architecture.



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2. Core Technology Deep-Dive

React.js 18 (Frontend) — UI Layer

- **Component Architecture:** Modular hierarchy (Cards, Filters, Grids).
- **Virtual DOM:** Sub-16ms smooth 60fps re-renders during filtering.
- **Tailwind CSS:** Zero-runtime CSS with utility-first styling.
- **Edge CDN:** Deployed on Vercel with automatic SSL and caching.

FastAPI (Backend) — API Layer

- **Async Execution:** Built on Starlette and Uvicorn ASGI server.
- **Auto Documentation:** Interactive OpenAPI / Swagger UI at /docs.
- **Pydantic v2:** Strict type validation with Rust core speed.
- **JWT Security:** Cryptographically signed auth tokens.

PostgreSQL 15 (DB) — Data Layer

- **ACID Guarantee:** Zero data corruption for user accounts.
- **JSONB Support:** Efficient native storage of dynamic 10-day plans.
- **Connection Pool:** Scalable connection recycling via SQLAlchemy.
- **Relational Integrity:** Foreign key cascades between users and trips.

3. Technology Selection Trade-Off Analysis

STACK LAYER	SELECTED TECH	ALTERNATIVES EVALUATED	DECISIVE ENGINEERING RATIONALE
Frontend	React.js 18	Vue.js, Angular, Svelte	Largest ecosystem, seamless Vercel integration, superior component reusability.
Backend	FastAPI	Django, Flask, Node.js	3x higher throughput than Flask, built-in async support, automatic OpenAPI schemas.
Database	PostgreSQL 15	MongoDB, MySQL, SQLite	Best-of-both-worlds: relational ACID security for users + JSONB for flexible day plans.
Hosting	Vercel CDN	AWS EC2, Heroku, DigitalOcean	Global edge caching, zero maintenance, instant deployment via Git webhook pushes.